ASSEMBLY GUIDE – DESK BUDDY V1.0

1. PRINTING THE COMPONENTS

All components can be printed in PLA or PETG without using supports and in any colors you prefer. The attached .3mf file already includes the optimal print orientations for all components.

2. REQUIRED COMPONENTS

The following components are required to build the project:

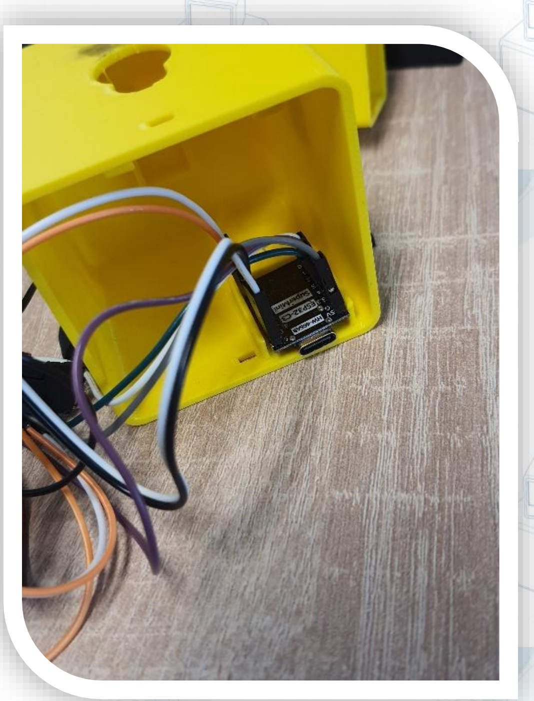
- ESP C3 Super Mini
- SG90 servomotor
- SSD1306 display
- Touch sensor (TTP223 capacitive touch sensor)
- Jumper wires
- USB C Cable (for programming the ESP32)

3. ASSEMBLY

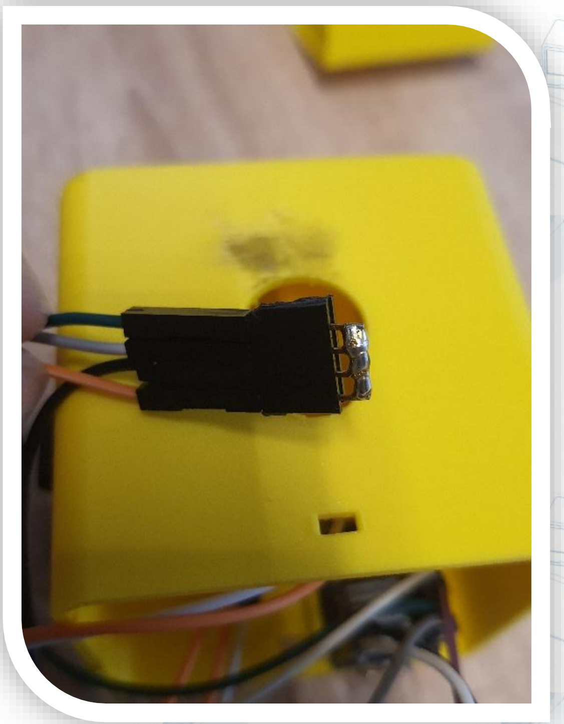
Once all parts have been printed, follow the steps below to assemble the robot. Make sure you have already completed the wiring between the various components before starting the assembly procedure, except for the display.

- Glue the servomotor horn to the robot's head using hot glue or super glue, trying to keep it aligned with the guide built into the model.

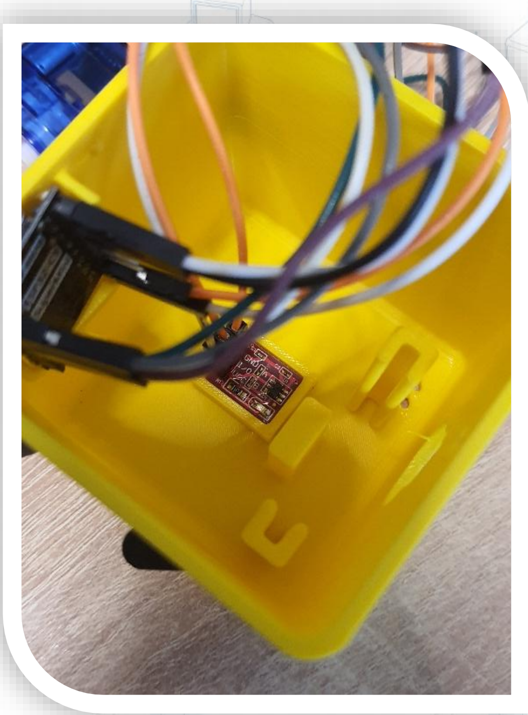
- Place the ESP32 C3 Super Mini into its dedicated housing and secure it with hot glue or double-sided tape.



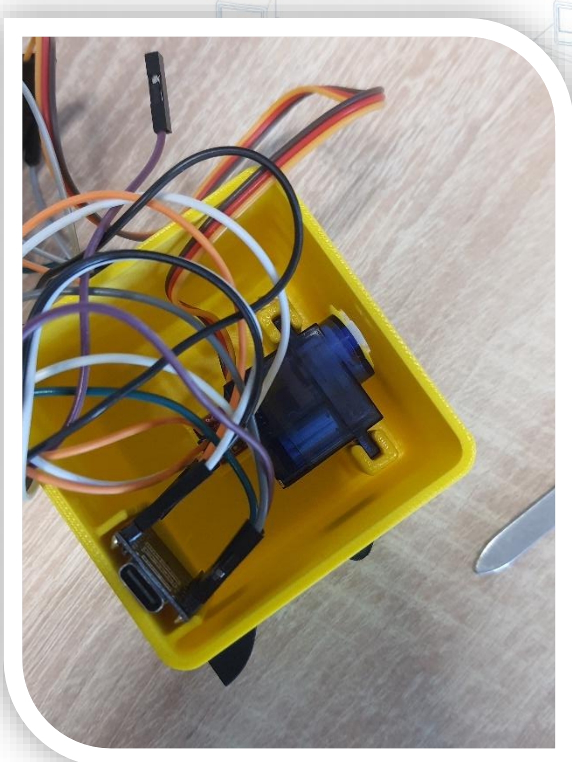
- Prepare some soldered pins or a terminal block to create a common GND connection, which will help connect everything to the single GND pin on the ESP32.



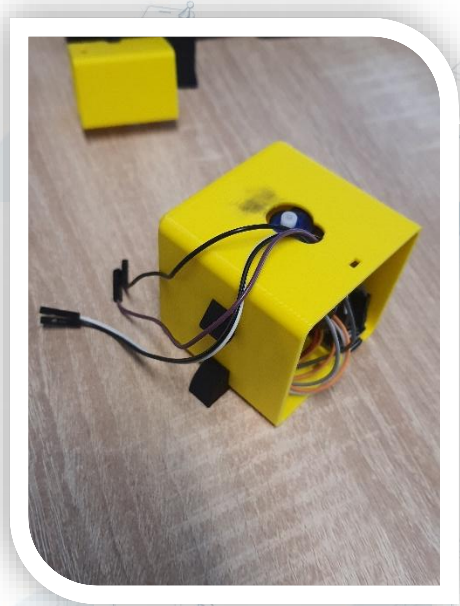
- Place the TTP223 into its dedicated housing and, if necessary, apply light pressure with pliers or a tool to snap it into place.



- Insert the SG90 motor as shown in the image. Please note that there is only one correct orientation, as shown in the figure. Make sure the servo shaft is exposed at the top of the robot body and passes through the central hole.



- Pass the wires needed to connect the display through the central hole as well.



- Position the display cover, then pass the wires inside the head and connect them to the display before pushing it into the correct position.

- Apply the rear display cover to secure everything in place.

- Tidy up the wires inside the main body and properly retract the wires inside the head to prevent bulges or friction with the motor movement. Close everything by positioning the rear body cover, paying attention to the opening required for the USB-C port connection.

4. STARTUP AND PROGRAM UPLOAD

You can use MicroPython on the ESP32 C3 Super Mini. Follow these steps to get the robot working:

• Update the firmware on the ESP32 C3 Super Mini by following this guide: https://micropython.org/download/ESP32_GENERIC_C3/
• Download the contents of the GitHub repository https://github.com/MarkOmez/DeskBuddyRobot and, using a MicroPython editor such as Arduino Lab for MicroPython (https://labs.arduino.cc/en/labs/micropython), upload the files to the board.
• Have fun using your robot.

5. USAGE

The robot has a series of built in action that can be triggered by pressing the capacitive button. By default animated eyes will be displayed on startup and it is possible, by a single brief press to see the actual time, today and tomorrow weather (make sure to set the variables in the code).

With a long press is also possible to enable or disable the movement of the head.

6. LIABILITY DISCLAIMER

The assembly instructions provided are intended for informational and descriptive purposes only. The author accepts no responsibility for any damage, malfunction, or failure of equipment, electronic components, or other devices resulting from carrying out the procedure described.

The assembly, wiring, and use of the components must be carried out carefully and under the user's full responsibility. Always check the wiring, supply voltages, and component compatibility before powering on the device.